

```
pip install tensorflow
```

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Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.19.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.4.0)
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Requirement already satisfied: protobuf!=4.21.0,!=4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<6.0.0dev,>=3.20.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (4.25.1)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.32.4)
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Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.3.0)
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```
pip install tensorflow-datasets
```

```
Requirement already satisfied: tensorflow-datasets in /usr/local/lib/python3.12/dist-packages (4.9.9)
Requirement already satisfied: absl-py in /usr/local/lib/python3.12/dist-packages (from tensorflow-datasets) (1.4.0)
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Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages (from promise->tensorflow-datasets) (1.17.0)
Requirement already satisfied: docstring-parser~=0.15 in /usr/local/lib/python3.12/dist-packages (from simple_parsing->tensorflow-datasets) (0.16.1)
Requirement already satisfied: googleapis-common-protos<2,>=1.56.4 in /usr/local/lib/python3.12/dist-packages (from tensorflow-datasets) (1.66.0)
```

```
import tensorflow as tf
import tensorflow_datasets as tfds
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten, Dropout
from tensorflow.keras.utils import to_categorical
import matplotlib.pyplot as plt
import numpy as np
```

```
print("Loading EMNIST Letters dataset..")
(ds_train,ds_test), ds_info=tfds.load(
    'emnist/letters',
    split=['train','test'],
    shuffle_files=True,
    with_info=True,
)
def preprocess_emnist(features):
    image=tf.cast(features['image'],tf.float32)/255.0
    image=tf.image.transpose(image)
    label=features['label']-1
    return image,label

train_data=ds_train.map(preprocess_emnist).batch(32).prefetch(tf.data.AUTOTUNE)
test_data=ds_test.map(preprocess_emnist).batch(32).prefetch(tf.data.AUTOTUNE)
```

Loading EMNIST Letters dataset..

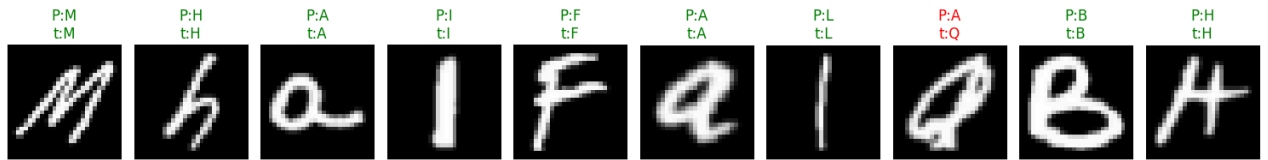
```
with tf.device('/GPU:0'):
    model=Sequential([
        Flatten(input_shape=(28,28,1)),
        Dense(256,activation='relu'),
        Dropout(0.2),
        Dense(128,activation='relu'),
        Dropout(0.2),
        Dense(26,activation='softmax')
    ])
```

/usr/local/lib/python3.12/dist-packages/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an `input\_shape` /
super().__init__(**kwargs)

```
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
print("\nStarting training on 26 letter classes...")
history=model.fit(
    train_data,
    epochs=10,
    validation_data=test_data
)
```

```
Starting training on 26 letter classes...
Epoch 1/10
2775/2775 ————— 50s 17ms/step - accuracy: 0.6351 - loss: 1.2264 - val_accuracy: 0.8453 - val_loss: 0.4887
Epoch 2/10
2775/2775 ————— 34s 12ms/step - accuracy: 0.8327 - loss: 0.5378 - val_accuracy: 0.8651 - val_loss: 0.4163
Epoch 3/10
2775/2775 ————— 35s 13ms/step - accuracy: 0.8562 - loss: 0.4474 - val_accuracy: 0.8725 - val_loss: 0.3815
Epoch 4/10
2775/2775 ————— 32s 9ms/step - accuracy: 0.8705 - loss: 0.4014 - val_accuracy: 0.8807 - val_loss: 0.3672
Epoch 5/10
2775/2775 ————— 21s 7ms/step - accuracy: 0.8757 - loss: 0.3806 - val_accuracy: 0.8822 - val_loss: 0.3609
Epoch 6/10
2775/2775 ————— 43s 8ms/step - accuracy: 0.8834 - loss: 0.3503 - val_accuracy: 0.8859 - val_loss: 0.3486
Epoch 7/10
2775/2775 ————— 23s 8ms/step - accuracy: 0.8870 - loss: 0.3358 - val_accuracy: 0.8882 - val_loss: 0.3454
Epoch 8/10
2775/2775 ————— 39s 14ms/step - accuracy: 0.8916 - loss: 0.3220 - val_accuracy: 0.8922 - val_loss: 0.3375
Epoch 9/10
2775/2775 ————— 35s 13ms/step - accuracy: 0.8950 - loss: 0.3129 - val_accuracy: 0.8900 - val_loss: 0.3407
Epoch 10/10
2775/2775 ————— 37s 13ms/step - accuracy: 0.8965 - loss: 0.3069 - val_accuracy: 0.8918 - val_loss: 0.3356
```

```
def get_letter(label):
    return chr(ord('A')+label)
images,labels= next(iter(test_data))
predictions=model.predict(images)
plt.figure(figsize=(15,4))
for i in range(10):
    plt.subplot(1,10,i+1)
    plt.imshow(images[i].numpy().squeeze(),cmap='gray')
    pred_idx=np.argmax(predictions[i])
    true_idx=labels[i].numpy()
    color='green' if pred_idx==true_idx else 'red'
    plt.title(f"P:{get_letter(pred_idx)}\nt:{get_letter(true_idx)}", color=color)
    plt.axis('off')
plt.tight_layout()
plt.show()
```

1/1  0s 167ms/step

```
loss,accuracy=model.evaluate(test_data,verbose=0)
print(f"\nFinal test accuracy for letters:{accuracy*100:.2f}%")
```

Final test accuracy for letters:89.18%